

EUROPEAN ECONOMICS ANALYST

Euro Area—Introducing Our Nearcast Model for Inflation

- Energy prices remain highly volatile and, despite generally benign spot data since the start of the Middle East crisis, core inflation forecasts for 2027 have continued to drift up. Against this backdrop, we launch a new bottom-up Euro area inflation model that incorporates information from over 600 domestic and global price-related variables and is thus able to provide an updated near-term inflation forecast (or “nearcast”) on a daily basis.
- Our model consists of three steps. First, we include all non-HICP variables, i.e., surveys, commodity forwards, and hard price data, in a dynamic factor model (DFM) of cost pressures. The underlying factors are intended to capture signals on specific upstream production costs, such as wages, energy costs, and other physical inputs prices. The estimated cost factors (a) summarise the information from several noisy global and domestic price-related variables, and (b) can be projected forward with the cross-product linkages allowed by the restricted transition matrix of the DFM.
- Second, we leverage input-output tables to compute HICP component-specific cost indices as weighted averages of the upstream cost pressures estimated in the DFM. Third, we combine the resulting component-level cost indices and other domestic price indicators in statistical models (such as VARs) to forecast component-level HICP inflation. We can then aggregate these into a forecast for Euro area core and headline inflation that can be refreshed on a daily basis as new input data are released.
- Our bottom-up model results in large out-of-sample forecast gains relative to simple benchmarks for all major components at the 3-, 6-, and 12-month horizon. More importantly, a pseudo real-time exercise suggests that the model would have performed better than other forecasts in 2022 by signalling inflation upside very early in the year. Our model performed broadly in line with other forecasts in 2023-25.
- Over 2026, the model-implied headline inflation rate for end-2026 has risen sharply from below 2%yoy as of February to just under 4% as of now. Core inflation is currently projected to settle a touch above 2.6%yoy in December, up from 2% expected prior to the start of the Middle East war. Implied core pressures rose in H1, eased slightly in July, and have firmed up again since August.

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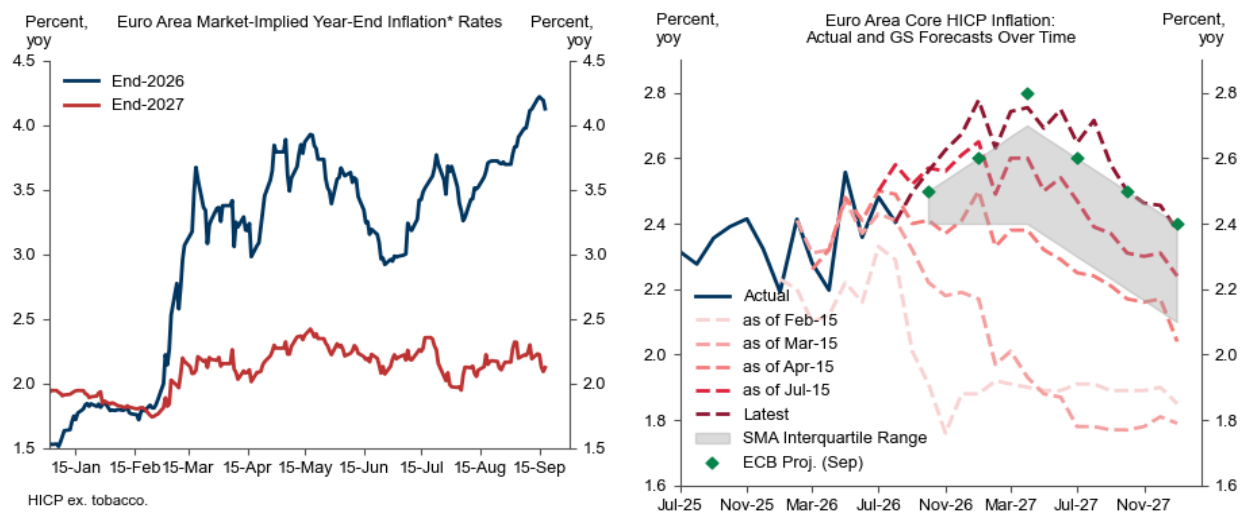
- Our new model is currently close to our recently updated inflation forecasts but signals some upside risk to core goods pressures. We will run our nearcast model on a daily basis going forward and report the resulting projections in our daily *Euro Area Data Update*.

Euro Area—Introducing Our Nearcast Model for Inflation

Since the start of the Middle East crisis, energy prices have remained highly volatile. The recent moves in oil, refined products, and natural gas prices have resulted in a new rise in market-implied inflation, with contracts currently implying HICP ex. tobacco inflation above 4%yoy in December ([Exhibit 1](#), left).

At the same time, spot core inflation data has continued to look fairly benign. HICP inflation ex. food, energy, alcohol and tobacco printed at 2.40%yoy in August, broadly the same rate observed in February and only a 0.1pp above our forecast for August core inflation as of prior to the start of the conflict ([Exhibit 1](#), left). Despite this, our and consensus forecasts and the recent ECB staff projections continue to point to growing upside to core inflation from here, albeit with elevated uncertainty around the strength of indirect effects.

Exhibit 1: High Volatility in Commodity Prices and Uncertain Core Pass-Through



Source: Bloomberg, SMA, ECB, Haver Analytics, Goldman Sachs Global Investment Research

Interpreting the inflation-related dataflow in the Euro area is quite challenging given the staggered nature of national and area-wide releases and the wide range of non-HICP indicators available (such as surveys, PPIs, import prices, and wholesale prices), which can frequently send noisy or conflicting signals.

Against this backdrop, we launch a new bottom-up inflation model that incorporates in a coherent framework over 600 domestic and global price-related variables. This rich dataset of inputs enables the model to track price pressures in the pipeline on a high-frequency basis by providing an updated model-implied forecast (or “nearcast”) every time new price-relevant data are released.

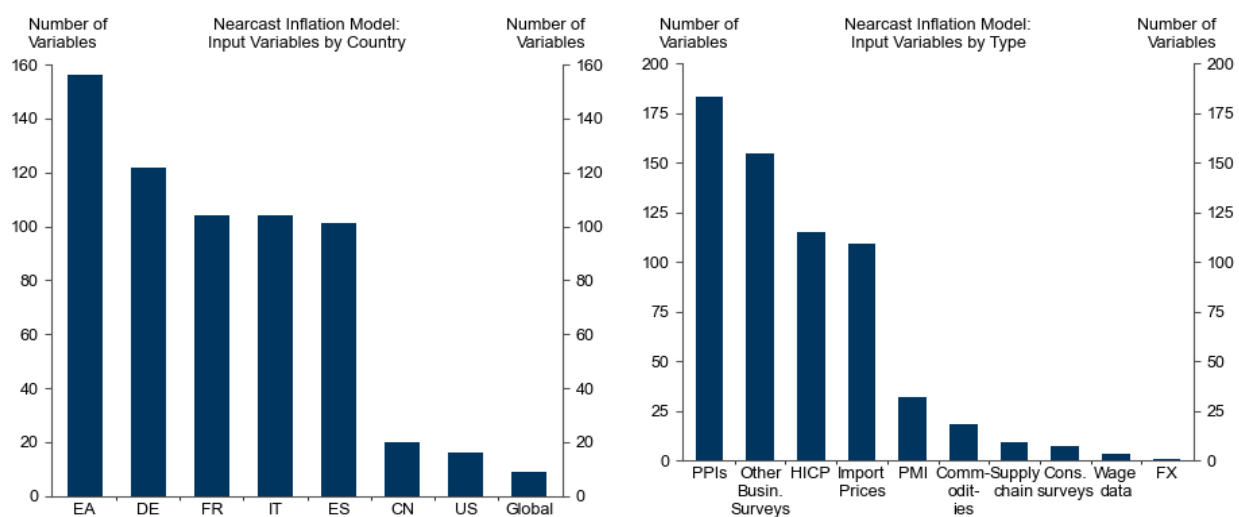
A New High-Frequency Inflation Model for Near-Term Inflation

A key advantage of the new model is the wealth of information it considers ([Exhibit 2](#)). Inputs include:

- 23 HICP subindices for the Euro area and each of the EMU4 economies;

- Sectoral PPIs for the Euro area, the EMU4, China, and the US;
- Component-level German wholesale prices;
- Import prices for the Euro area and the EMU4;
- Sectoral selling price expectations from the EC survey and national non-PMI surveys;
- Aggregate and sectoral PMI price indices;
- Area-wide compensation per employee, wage trackers, consumer confidence, and inflation expectations;
- Survey-based supply chain indicators;
- Daily and weekly prices for Brent, TTF, and refined products (with forwards where available);
- Daily and monthly non-energy commodity prices (such as fertilisers and metals prices);
- The ECB's Euro trade-weighted index.

Exhibit 2: Our Model Considers Over 600 Domestic and Global Price-Related Variables



Source: Goldman Sachs Global Investment Research, Haver Analytics

While our framework leverages a large amount of country-level information, the targets for our nearcast model are 23 exhaustive components of the area-wide HICP basket, representing an ad hoc breakdown of the basket sitting roughly between the three- and four-digit COICOP groups.

Our nearcast model consists of three steps.

#1 An Underlying Model of Upstream Cost Pressures

The first bloc of the model is a large dynamic factor model (DFM), a statistical tool to reduce a large number of variables into a smaller set of common factors that capture the underlying co-movements and trends in the data. In our model, the underlying factors are designed to capture various production cost pressures.

The DFM is fed a large dataset of all non-HICP variables in our model that includes sectoral EMU4 and foreign PPIs, sectoral import prices, wholesale prices, wage

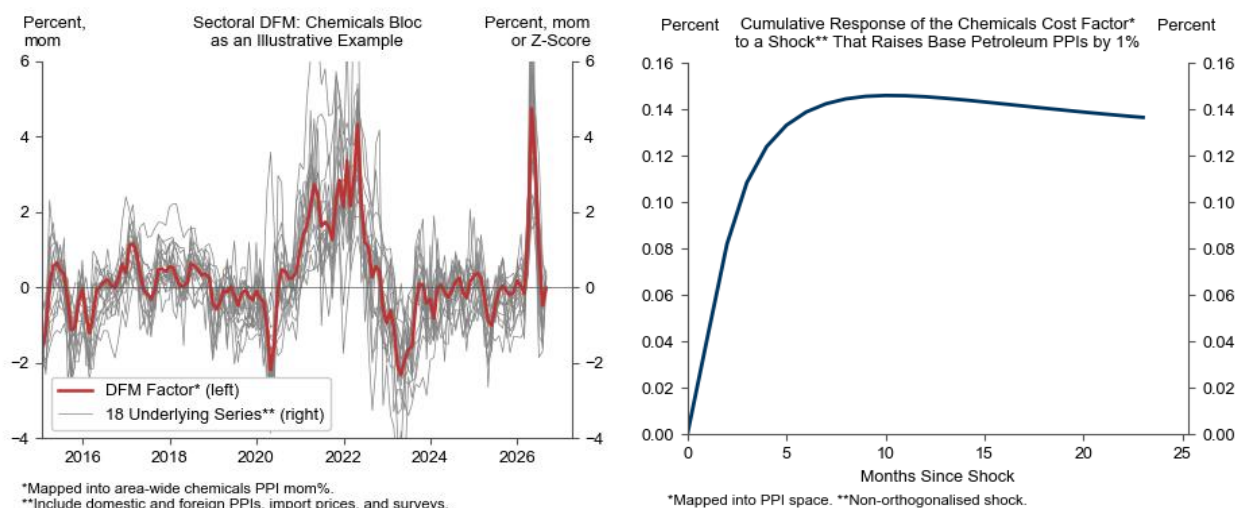
indicators, energy commodity prices, refined product prices, supply-chain pressures indicators, and the ECB trade-weighted EUR index. The DMF returns a set of estimated underlying cost factors. A detailed explanation of the DFM is available in the Appendix.

The DFM serves two purposes.

First, the extracted common factors express the common signal from the several noisy indicators available in each sector. For instance, a single time series will summarise the information from all EMU4 chemicals PPIs and import prices, wholesale prices, foreign chemicals PPIs, and survey indicators for the chemicals sector. This common signal is updated every time a new variable is released ([Exhibit 3, left](#)).

Second, it generates a forecast for these cost factors via its transition matrix. We restrict the transition matrix to allow only for economically sensible cross-sector spillovers. For example, the forward path of the extracted chemicals cost factor is allowed to be informed by past moves in oil prices. Contrarily, we do not allow our wage cost factor to respond to cost pressures in the paper sector, for instance ([Exhibit 3, right](#)).

Exhibit 3: An Underlying DFM Summarises Sectoral Cost Pressures



Source: Haver Analytics, Goldman Sachs Global Investment Research

#2 Combining Cost Factors in Component-Level Cost Indices

The DFM returns estimated cost-pressure factors (inclusive of a near-term forecast). To leverage these for inflation nearcasting, we need to map them into HICP components.

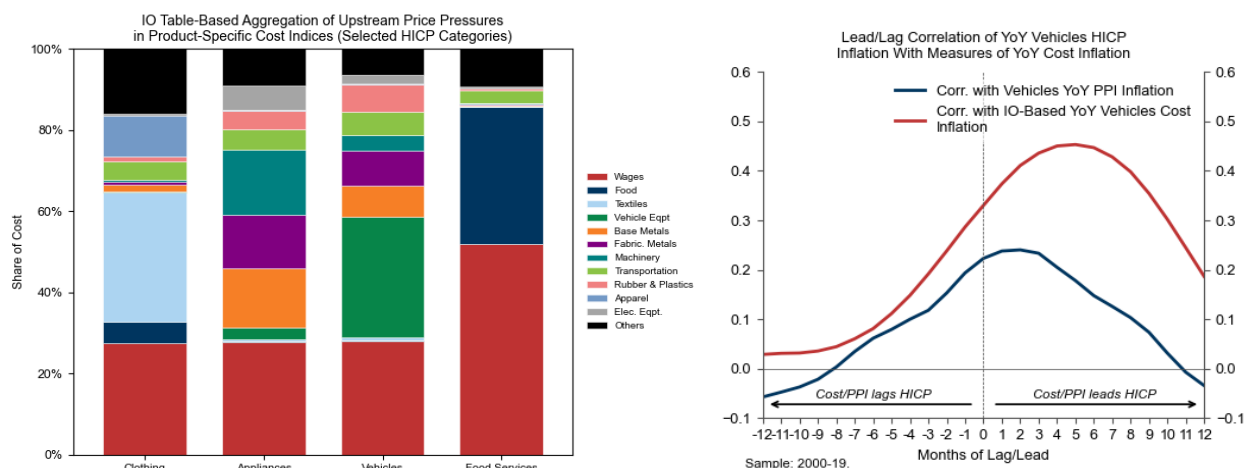
To do so, we leverage input-output (IO) tables, which describe the inputs needed to produce a given good or service consumer by household. The IO tables, for example, suggest that restaurants combine labour and food inputs to provide food services consumed in the Euro area. For clothing retailers, on the other hand, textiles costs are particularly relevant while labour costs are less significant in the overall input bill ([Exhibit 4, left](#)).

We use these direct cost shares to aggregate upstream costs (including labour and non-labour costs) into product-level cost indices that are specific to a given HICP component.

The resulting component-level cost indices provide a good leading signal for the

corresponding component's inflation rate, generally performing better than less detailed (and more noisy) alternatives such as raw PPIs ([Exhibit 4, right](#)).

Exhibit 4: We Aggregate Upstream Cost Pressures in Product-Specific Cost Indices Via Input-Output Tables



Source: Exiobase, Haver Analytics, Eurostat, Goldman Sachs Global Investment Research

#3 Forecasting Component-Level Inflation

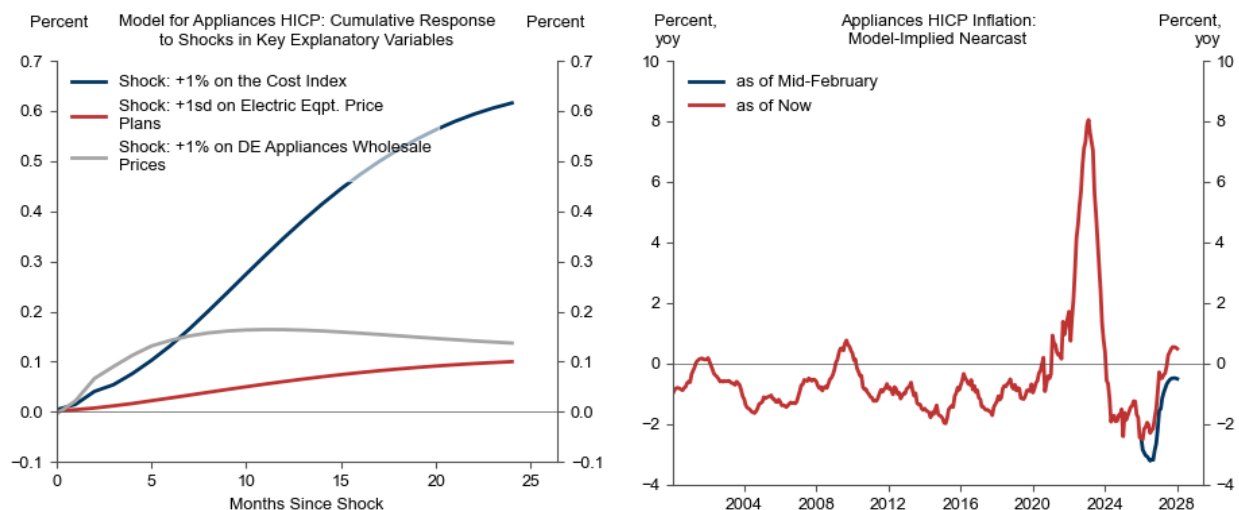
As a third step, we combine the component-level cost indices and other domestic price indicators in statistical models (mostly VARs and linear regressions) to forecast near-term component-level HICP inflation.

The exact specification used varies across components.

Fuels inflation, for instance, is largely forecast using oil futures to project forward refined product prices, assuming that product spreads narrow in line with the historical norm. For gas and electricity, we estimate a slow and gradual pass-through of wholesale natural gas prices, proxying for the contractual arrangements of consumer utilities. By using commodity forwards, our high-frequency model differs from our standing inflation forecasts, which are conditioned on our commodity strategists' forecasts.

For most core components, we leverage our component-level cost indices, frequently adding supplementary variables that have some explanatory power on top of the cost-indices, such as producers' or retailers' surveyed price expectations or some wholesale prices ([Exhibit 5](#), left shows the IRF from the VAR forecasting appliances inflation, as an example). In some instances (e.g., rent) we include past headline inflation in our forecasting models to proxy for catch-up effects.

While we target area-wide inflation rates, our models start nowcasting the area-wide figure as soon as an EMU4 economy has reported inflation details to ensure we leverage all information available.

Exhibit 5: We Combine Product-Specific Cost Indices with Other Variables to Forecast Inflation

Source: Haver Analytics, Eurostat, Goldman Sachs Global Investment Research

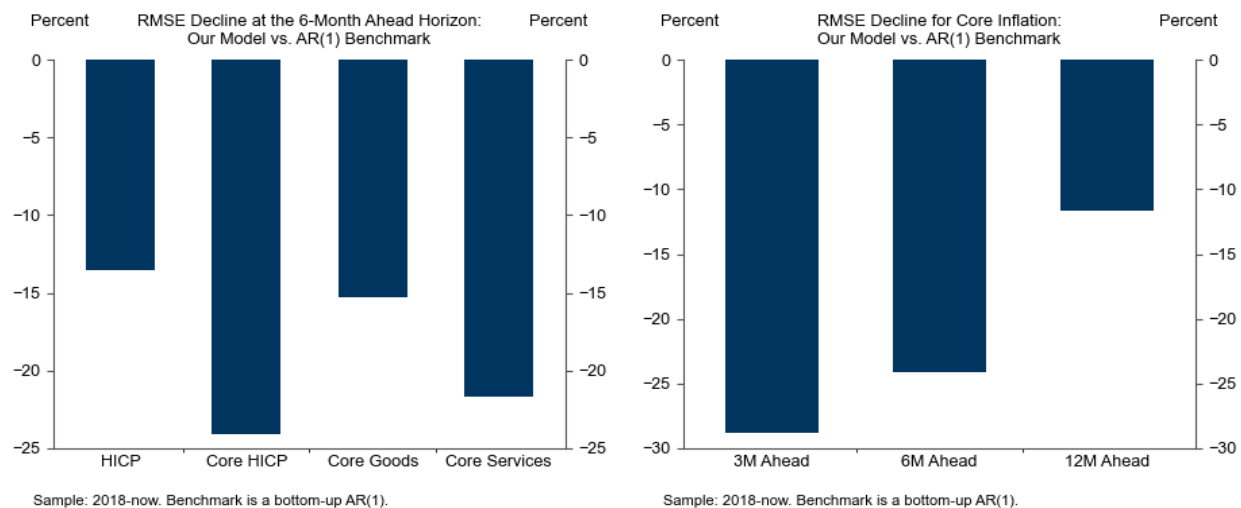
Our models can be refreshed on a daily basis as new input data are released, enabling us to assess changes in the inflation outlook implied by the dataflow ([Exhibit 5](#), right). The component-level forecasts can be aggregated up into core and headline forecasts.

Measuring Performance

We turn to assessing our nearcast model's performance. We do so in a pseudo real-time exercise, where we obtain an "as-of" input dataset based on average release lags. Our forecasting models are re-estimated at every run.¹

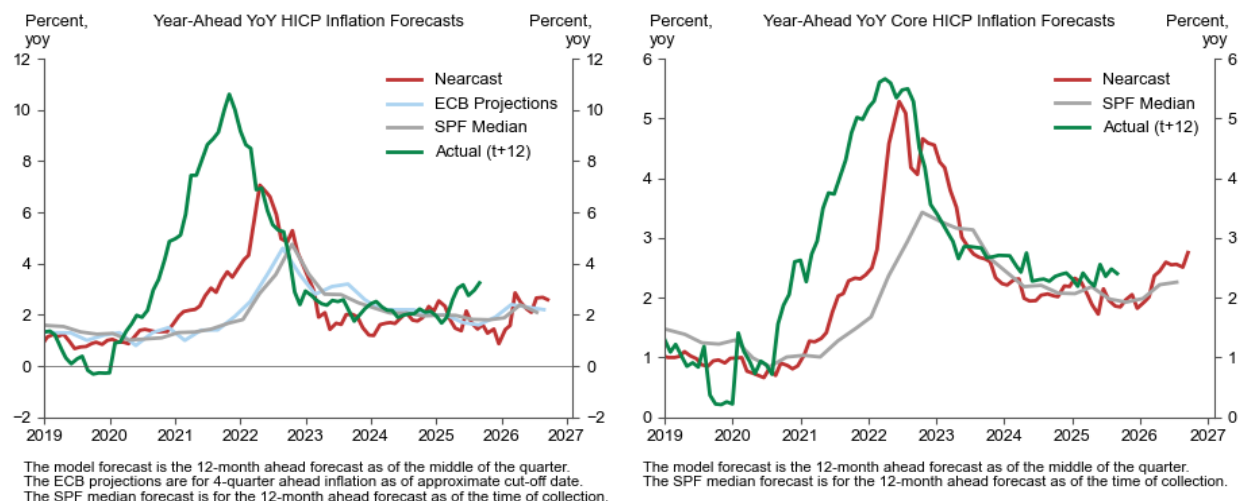
Our model results in large out-of-sample forecast gains relative to simple benchmarks for all major components. Forecasting gains are larger for core than headline, reflecting elevated forecasting errors for energy inflation in the sample considered. Our high-frequency model for core inflation outperforms the benchmark at the 3-, 6-, and 12-month horizon ([Exhibit 6](#)).

¹ At every run of the model on a vintage basis, we feed the as-of data to the DFM estimated over the full sample (which essentially keeps the relationship between cost factors fixed). All other components of the model, including the VARs and linear regressions used in forecasting, are re-estimated every time.

Exhibit 6: Meaningful Forecast Improvements 3- to 12-Months Ahead

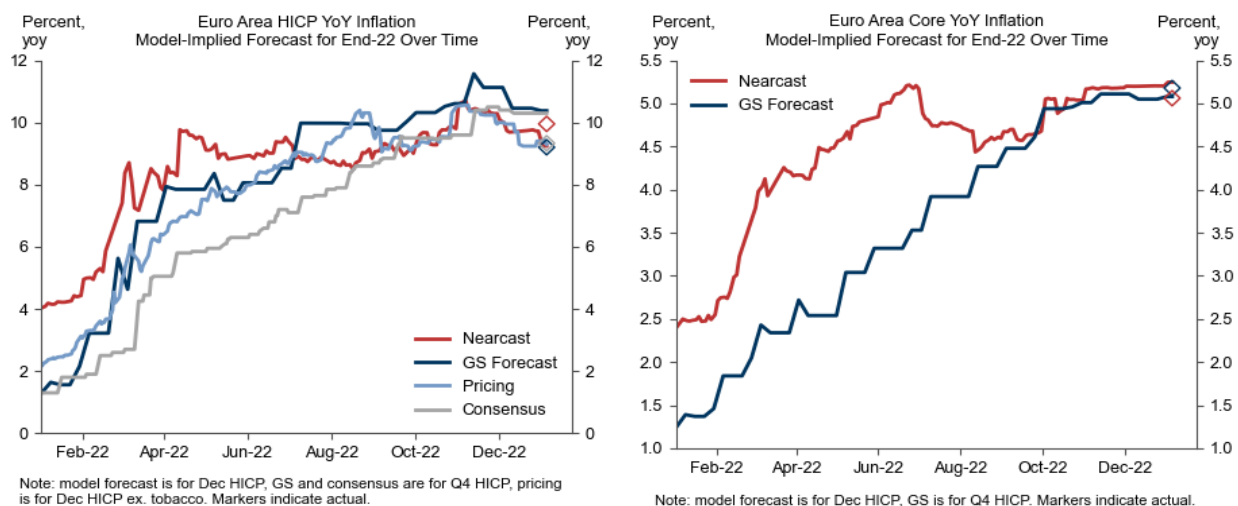
Source: Haver Analytics, Goldman Sachs Global Investment Research

We next assess the out-of-sample performance of our nearcast model against “smarter” benchmarks. For both headline and core, our high-frequency model would have performed better than the ECB projections and the median forecast from the Survey of Professional Forecasters (SPF) in the 2022 cycle. Since 2023, our model has performed broadly in line with other forecasts ([Exhibit 7](#)).

Exhibit 7: A Good Track Record

Source: Haver Analytics, SPF, ECB, Goldman Sachs Global Investment Research

Zooming into 2022, our nearcast model would have started the year with an end-year projection for headline and core inflation above other forecasts. The model-implied inflation trajectory would have risen further quite quickly after the start of the energy crisis. The nearcast would have been around 9% and 5% respectively for headline and core (close to the values eventually realised in December) by the middle of the year ([Exhibit 8](#)).

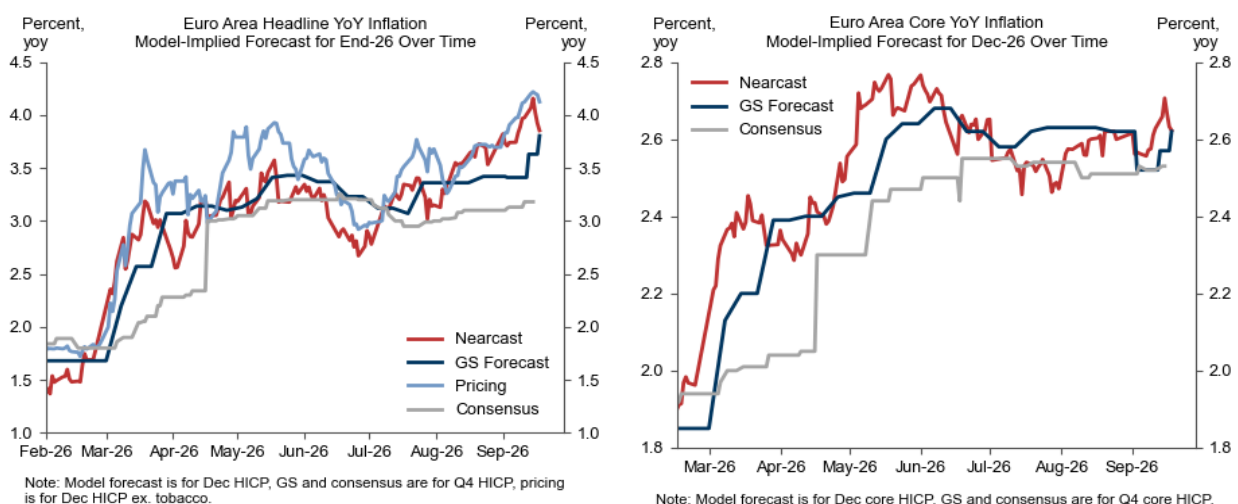
Exhibit 8: Our Model Would Have Picked Up Inflationary Pressures Building Early in 2022

Source: Haver Analytics, Bloomberg, Goldman Sachs Global Investment Research

The Current Signal

Over the course of this year, the model-implied headline inflation rate for end-2026 has risen sharply from below 2%yoy as of February to just under 4% now. The model-implied forecast has broadly followed commodity prices and has tracked market-implied inflation reasonably well. The model headline forecast currently sits at 3.85%yoy, below market pricing and close to our current official forecast ([Exhibit 9, left](#)).

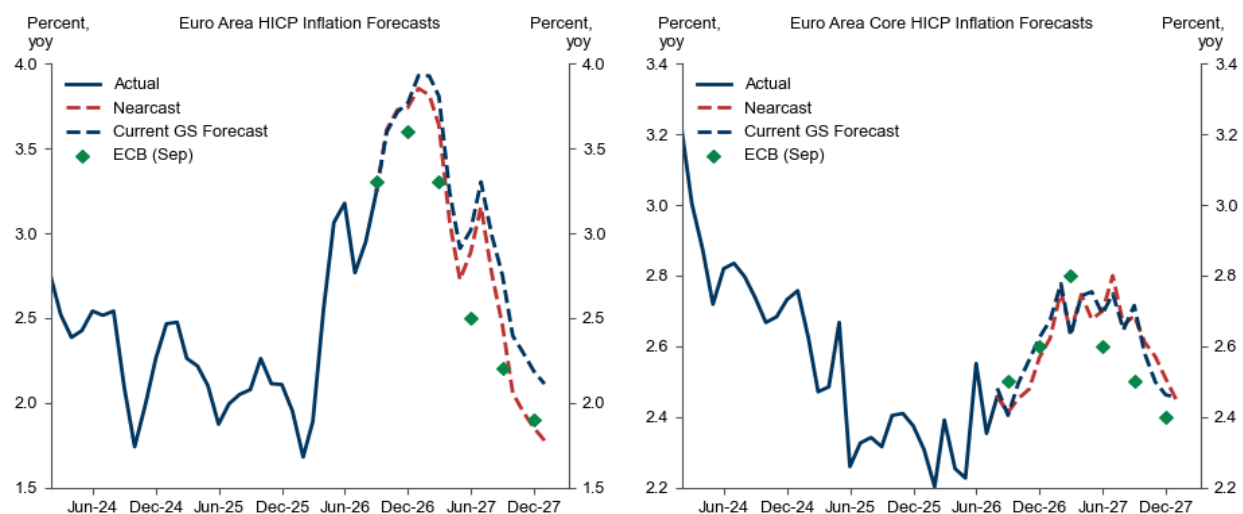
Core inflation for end-2026 is currently projected to settle at 2.62%yoy, up from around 2% expected prior to the start of the war. Implied core pressures rose in H1 on the back of rising energy prices and hawkish readings from manufacturing price surveys, supply-chain indices, and PPIs. Model-implied pressures eased slightly at the start of the summer but have firmed up again since August as energy commodity prices rallied further and non-HICP prices turned more hawkish again ([Exhibit 9, right](#)).

Exhibit 9: A Further Rise in Model-Implied Pressures Since August

Source: Haver Analytics, Bloomberg, Goldman Sachs Global Investment Research

Following recent updates, our standing inflation forecast is now quite close to the path implied by the nearcast model, signalling end-year headline just under 4%yoy and a peak of core inflation at 2.8% early next year (Exhibit 10).

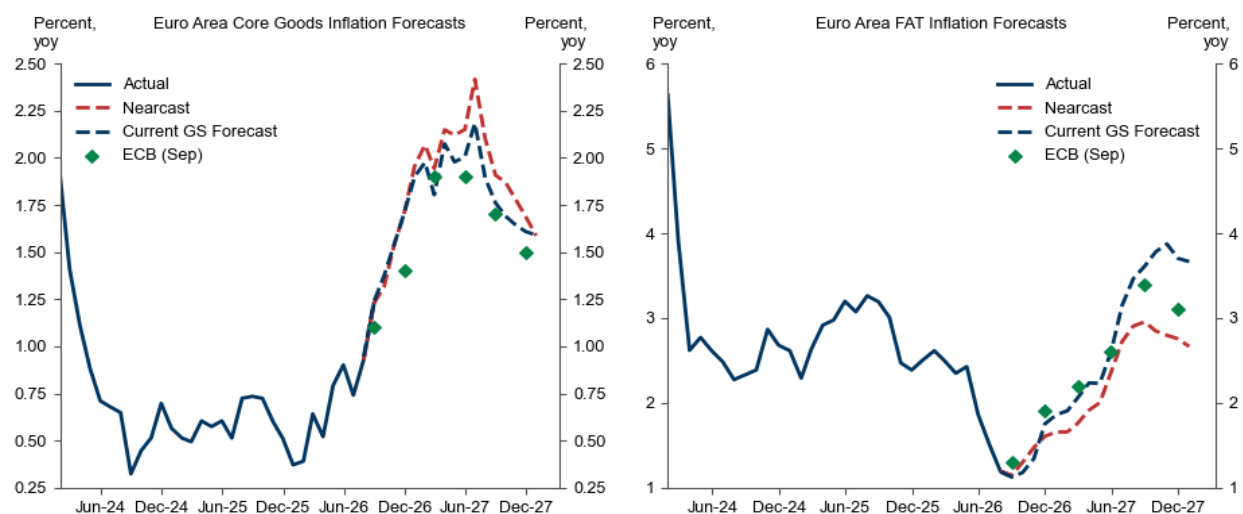
Exhibit 10: Somewhat Persistent Core Inflation...



Source: Haver Analytics, Goldman Sachs Global Investment Research, ECB

Across major categories, our high-frequency model signals upside risks relative to our and the ECB's latest forecasts as largely arising from core goods. Across more granular components, upside seems to be arising from vehicles, appliances, and ICT equipment. To the downside, our high-frequency model sees a more modest uptick in food inflation than currently in our and the ECB's forecasts, with a peak just over 3%yoy (Exhibit 11).

Exhibit 11: ... Largely Driven by Core Goods



Source: Haver Analytics, ECB, Goldman Sachs Global Investment Research

Moving forward, we will start running our high-frequency model on a daily basis and report the resulting projections in our daily *Euro Area Data Update*. The nearcast model will live alongside our official forecast – which is conditioned upon our commodity team's forecasts and incorporates our judgment – to interpret the broad inflation-relevant dataflow and express risks around our official forecasts.

Giovanni Pierdomenico**Appendix**

Our cost-pressure DFM is estimated on a dataset of all non-HICP variables in the model, largely standardised in mom% or z-score terms. These include EMU4 PPIs, import prices, selling price surveys, PMI price components, supply-chain indicators, commodity prices (forward filled with forwards where available), and select wage/services indicators.

Observables are grouped into cost blocs. Most of these reflect broad industrial sectors (petroleum, metals, food, chemicals, paper/wood, machinery, etc). We also include a factor for wage-related variables, one for transportation, one for repair work, and one for FX. These factors are jointly modelled with a restricted transition matrix with two lags which allows for spillovers.

In addition, we have country factors (one for each EMU4 economy, plus China and the US), a factor that loads on all import price indices, a “Soft” factor that loads on all surveys, and a “General” factor that loads on all domestic variables, including those that are not sector-specific (such as composite output prices in the PMIs).

We map the estimated cost factors into mom% changes of closely-related measure of price pressures for interpretability. For instance, the chemicals cost factor is mapped into area-wide chemicals PPI mom%, while the wage factor is mapped into compensation per employee.

Our cost factors are then aggregated into component-level cost indices using direct cost-contribution from the Exiobase input-output table.

We leverage the component-level cost indices in our bottom-up nearcast. Each component is forecast with an ad hoc model, largely selected on the basis of out-of-sample performance. Models are generally VARs or linear regressions that include the cost index and, on a case-by-case basis, additional explanatory variables such as PPIs, selling price expectations, lagged headline inflation (to proxy for catch-up effects), and FX moves. In our nearcasting models, inputs are for the Euro area as a whole but we nowcast them with national data whenever possible (e.g., we use national final details to nowcast component-level inflation at the area-wide level ahead of the final area-wide release).

The treatment of fuels is somewhat different. We use weekly pump prices from the European Commission and forecast those with oil forwards, assuming that products spreads converge to their long-run mean very gradually (according to a weekly AR process).

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